25901676 **Reg.No : .....**
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026
SECOND SEMESTER
Discipline Specific Core Course (DSC) - MG2DSCFSQ100 - FOOD AND NUTRITION
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
MCQ
Answer all questions
Each question carries **1** marks

1. Identify the RQ value for the complete oxidation of carbohydrate. [K] / [CO1, CO2]

a). 0.7 b). 0.8
c). 1 d). 1.2

2. Non essential amino acids : [U] / [CO1]

a). Must be obtained from the diet | b). Synthesized by the body | c). Not absorbed by the body | d). None of the above

3. One potential advantage of using biomarker monitoring in personalized nutrition [U] / [CO1, CO4]

a). It ensures every person eats the same diet | b). It provides real-time data to refine dietary recommendations | c). It eliminates the need for exercise | d). It focuses only on weight loss

4. A person who avoids dairy needs to meet their RDA for vitamin D. Which of these foods would be a good alternative? [U] / [CO1]

a). Oranges | b). Fortified plant-based milk | c). Potato chips | d). Butter

5. The condition which results when insufficient food is eaten over an extended period of tim [K] / [CO1]

a). Overnutrition | b). Under nutrition | c). Imbalance | d). Specific deficiency

6. Select the primary function of protein in the body [K] / [CO1]

a). Provide immediate energy | b). Build and repair tissues | c). Store fat | d). Increase water retention

7. Recognize which among the following is known as folic acid [U] / [CO1, CO3]

a). vitamin B1 b). vitamin B2 c). vitamin B3 d). vitamin B9

8. Identify from the following the following is the correct option as functions of fat [K] / [CO1]

a). aids in the absorption of fat soluble vitamins.
b). helps in production of hormone.
c). insulation of the body organ from excess heat.
d). All of the above.

9. Identify the vitamin deficiency that leads disease megaloblastic anemia [K] / [CO1]

a). Vitamin K b). vitamin B4 c). Vitamin E d). vitamin C

10. Calcium absorption in the body is enhanced by the presence of which vitamin? [K] / [CO1, CO3, CO4]

a). Vitamin A

b). Vitamin C

c). Vitamin D

d). Vitamin E

[1x10 = 10]

Part B

Short essay
Answer any **4** questions
Each question carries **5** marks

11. Give the definitions for nutrition and health [K] / [CO1]

12. Summarize the symptoms of iodine deficiency disorder (IDD) [U] / [CO1, CO3, CO4]

13. List the main food groups included in a plant-based diet. [K] / [CO1, CO4]

14. Explain on the methods of measurement of BMR. [U] / [CO1, CO2]

15. Write short notes on general symptoms of carbohydrate deficiency [U] / [CO1]

16. Discuss the role of macronutrients (proteins, carbohydrates, and fats) in space nutritio. [U] / [CO1, CO4]

[5x4 = 20]

Part C

Essay

Answer any **2** questions
Each question carries **10** marks

17.	Elaborate on Polysaccharides	[K] / [CO1]
18.	Explain the role of vitamin A in maintaining the overall health of a person.	[K] / [CO1]
19.	Explain the concept of climate-friendly nutrition and its importance in reducing environmental impact.	[U] / [CO1, CO4]
20.	Explain in detail about the sources, distribution and functions of water.	[U] / [CO1, CO2]

[10x2 = 20]